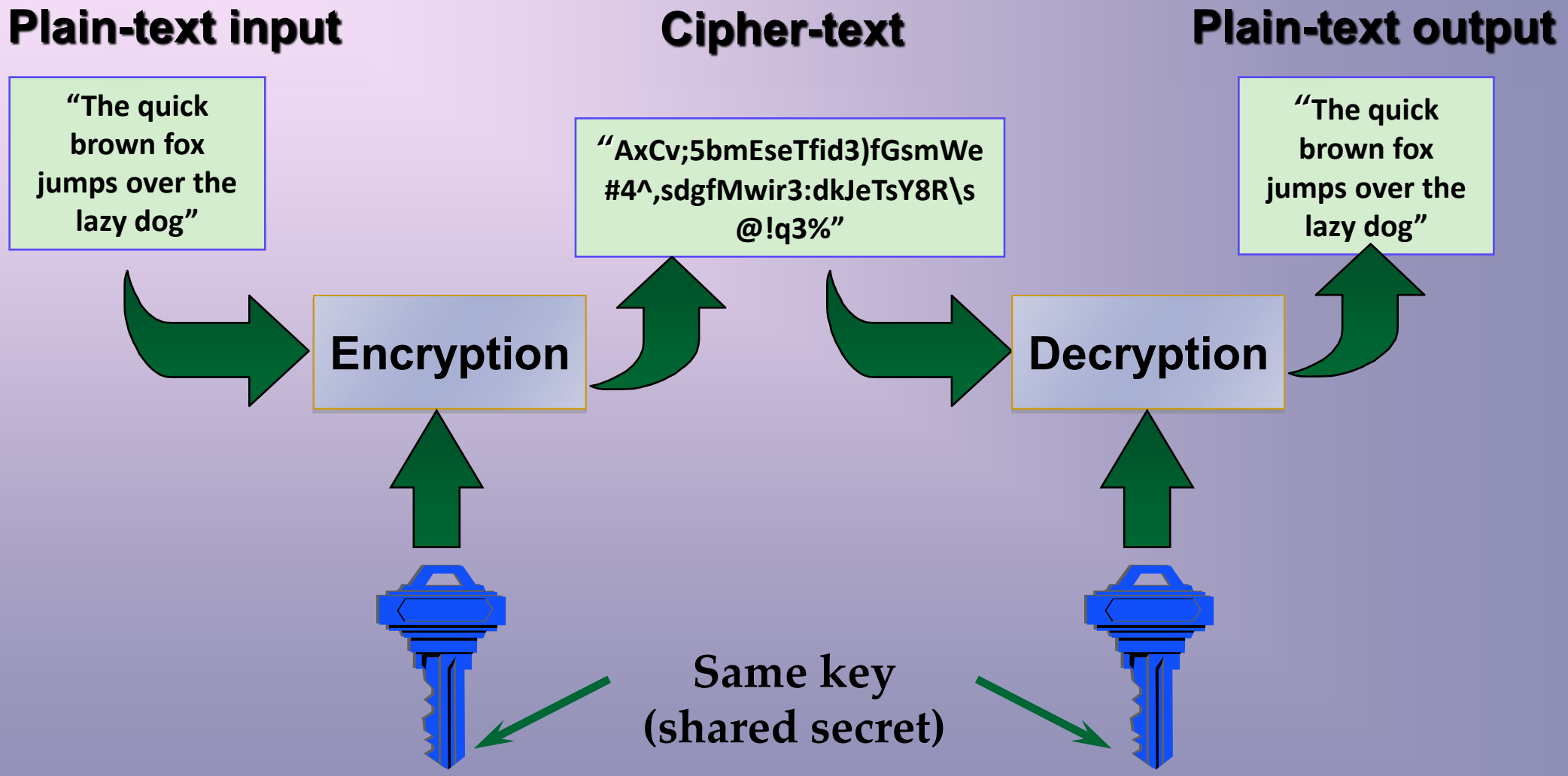
Public Key Cryptography

Symmetric Encryption

- Traditional **private/secret/single key** cryptography uses **one** key
  - shared by both sender and receiver
  - if this key is disclosed communications are compromised
  - also is **symmetric**, parties are equal
  - Disadvantage- it requires the prior communication of the *key* K between Alice and Bob.
- A secure channel is required
  - to let the key be known to the recipient
  - How to establish the secure channel – a practical problem
  - In practice, this may be very difficult to achieve.
  - Example- communication using email in between Alice and Bob (live far away from each other). In a situation such as this, Alice and Bob may not have access to a reasonable secure channel.

Symmetric Key Cryptography

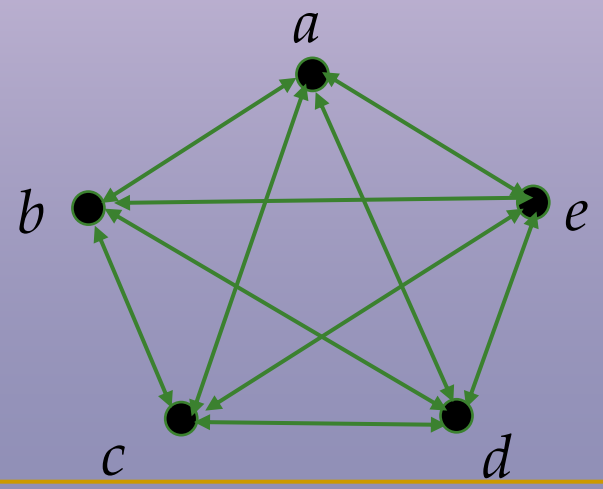


Symmetric Pros and Cons

- Strength:
  - Simple and really very fast (order of 1000 to 10000 faster than asymmetric mechanisms)
    - Super-fast (and somewhat more secure) if done in hardware (3DES, Rijndael)
- Weakness:
  - Must agree the key beforehand
  - Securely pass the key to the other party

Key Distribution Problem

- In symmetric key cryptosystems
  - Over complete graph with *n* nodes, $n(n-1)/2$ pairs secret keys are required.
  - Example: n=100, 99 x 50 = 4,950 keys are required
- Problem: Managing large number of keys and keeping them in a secure manner is difficult



Secret keys are required between

(a,b), (a,c), (a,d), (a,e), (b,c),

(b,d), (b,e), (c,d), (c,e), (d,e)

Public-Key Cryptography

- Probably most significant advance in the history of cryptography
- Uses **two** keys – a public & a private key
  - One for encryption and another one for decryption
  - Knowledge of the encryption key doesn't give knowledge of the decryptio*n* key
- **Asymmetric** since parties are **not** equal
- Uses number theoretic concepts to function

PKC motivation

- A cryptosystem where it is computationally infeasible to determine d_k given e_k.
- Encryption rule e_k is a *public key* which could be published in a directory
- Advantage- Alice (or anyone else) can send an encrypted message to Bob (without the prior communication of shared secret key) by using the public encryption rule e_k.
- Analogy-
  - 1. Alice places an object in a metal box, and then locks it with a combinational lock left there by Bob. Bob is the only person who can open the box since only he knows the combination.
  - 2. number of digits in a combinational lock, longer the key; stronger the encryption, as extending the length of the key exponentially increases the number of possible key combinations.

PKC motivation

- The idea of public-key cryptosystem was put forwarded by Diffee and Hellman in 1976.
- Then, in 1977,Rivest, Shamir, and Adleman invented the well-known RSA cryptosystem.

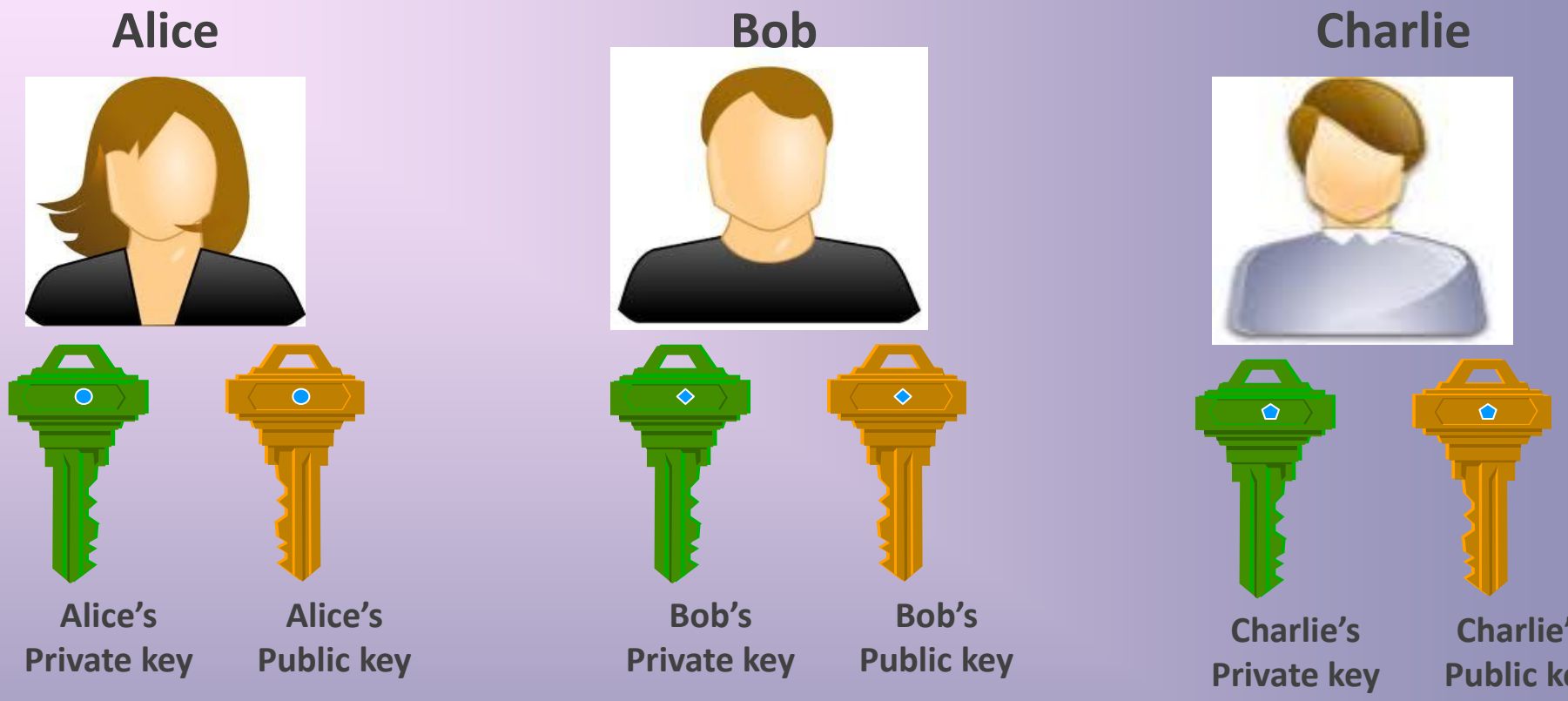
PKC motivation

- How many pairs of keys are required for say $\mathcal{N}$ users ?
- For N people to communicate there need only be N public/private key pairs.
- PKC was proposed as
  - communication over a public channel
    - using publicly known techniques

Public Key Cryptography

- developed to address two key issues:
  - key distribution – how to have secure communications in general without having to trust a KDC with your key
  - digital signatures – how to verify a message comes undamaged from the claimed sender

The Two Keys



- Each party has two keys

Main uses of Each Key

- A public-key
  - Public to anybody
  - used to encrypt messages and verify signatures
- A private-key
  - known only to the owner
  - used to decrypt messages, and sign (create) signatures

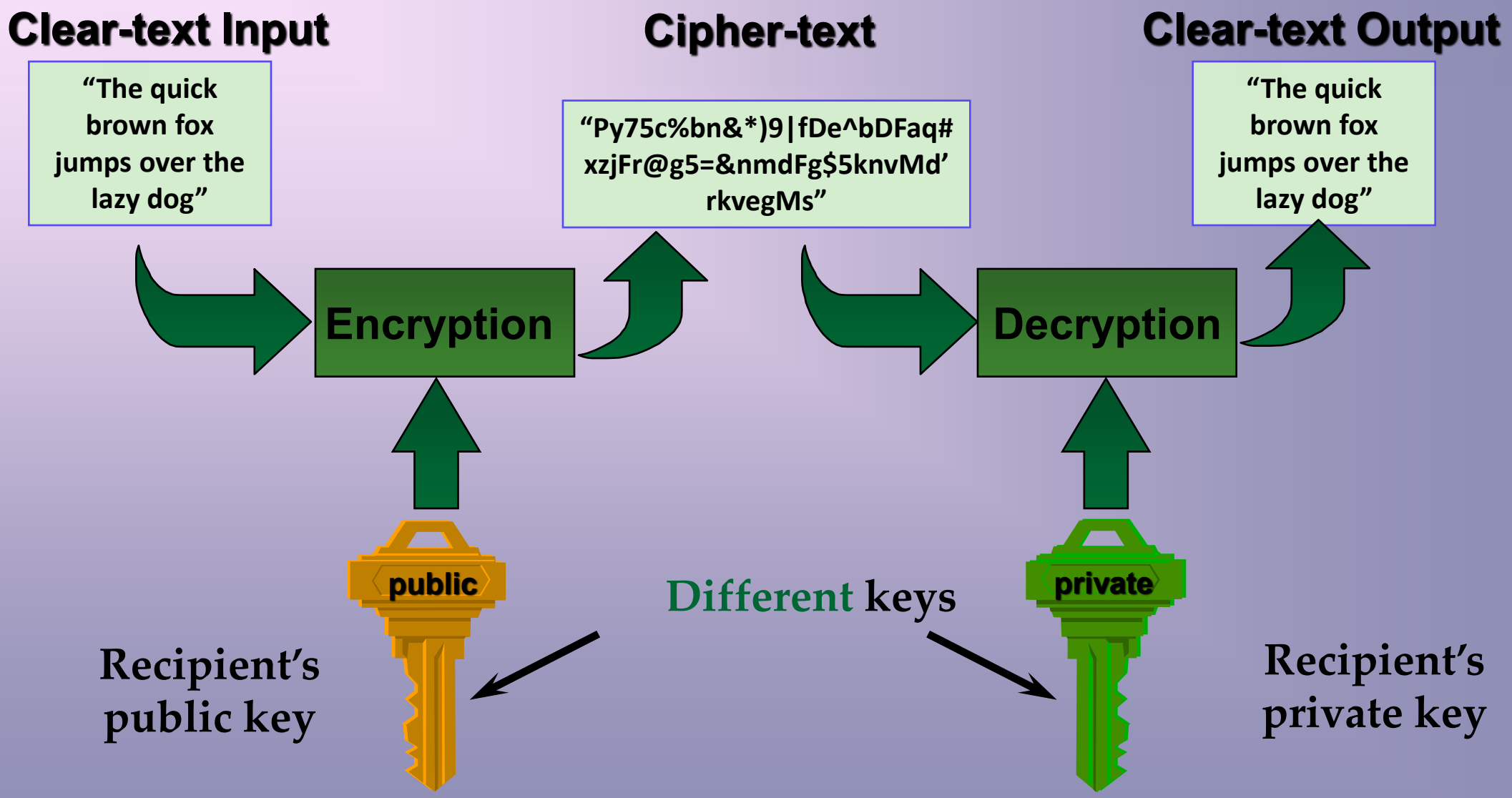
How does 2 different keys work?

- Just an very simple example:
  - Public Key = 4, Private Key = 1/4, message M = 5
  - **Encryption:**
    - Ciphertext C = M * Public Key
    - 5 * 4 = 20
  - **Decryption:**
    - Plaintext M = C * Private Key
    - 20 * ¼ = 5

An Example: Internet Commerce

- Bob wants to use his credit card to buy some brownies from Alice over the Internet.
- Alice sends her public key to Bob.
- Bob uses this key to encrypt his credit-card number and sends the encrypted number to Alice.
- Alice uses her private key to decrypt this message (and get Bob's credit-card number).

Public Key Encryption



Hybrid Encryption Systems

- All known public key encryption algorithms are much slower than the fastest secret-key algorithms.
- In a ***hybrid*** system, Alice uses Bob's public key to send him a secret shared ***session key***.
- Alice and Bob use the session key to exchange information.

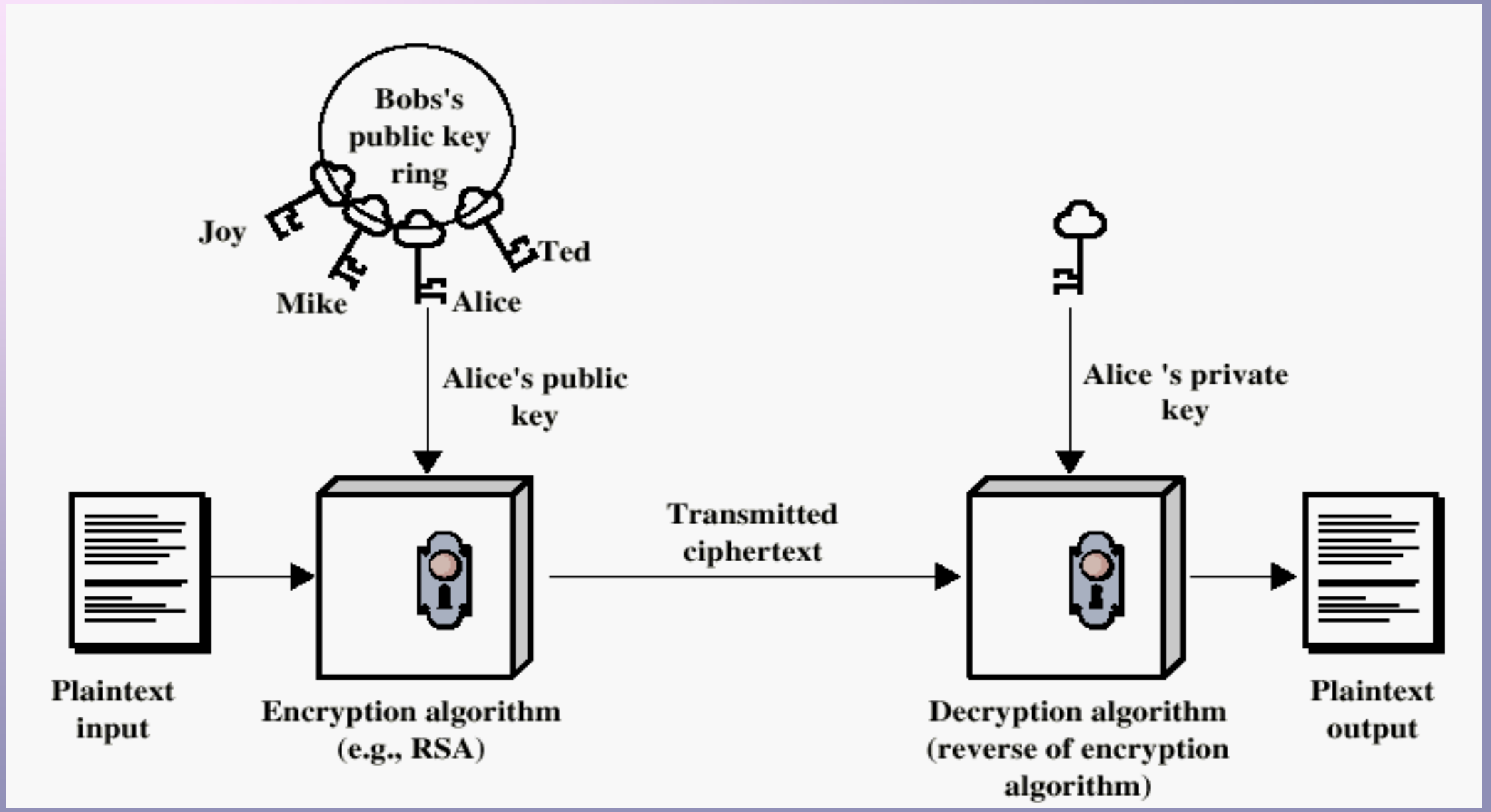
Public-Key Cryptography

- **public-key/two-key/asymmetric** cryptography involves the use of **two** keys:
  - a **public-key**, which may be known by anybody, and can be used to **encrypt messages**, and **verify signatures**
  - a **private-key**, known only to the recipient, used to **decrypt messages**, and **sign** (create) **signatures**
- is **asymmetric** because
  - those who encrypt messages or verify signatures **cannot** decrypt messages or create signatures

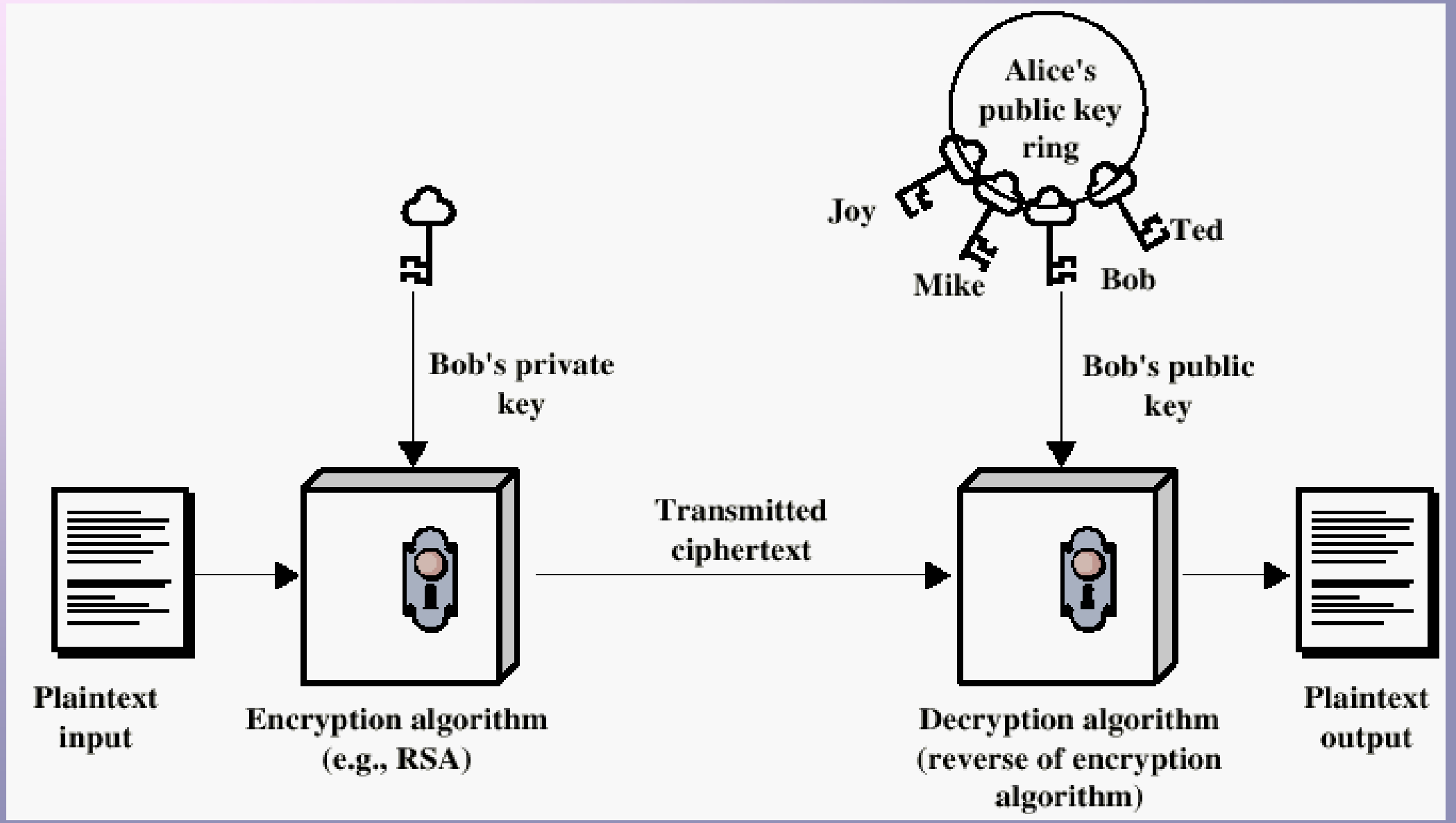
PKC Ingredients

- The scheme following ingredients
  - Plaintext
  - Encryption algorithm
  - Public and private key
  - Ciphertext
  - Decryption algorithm

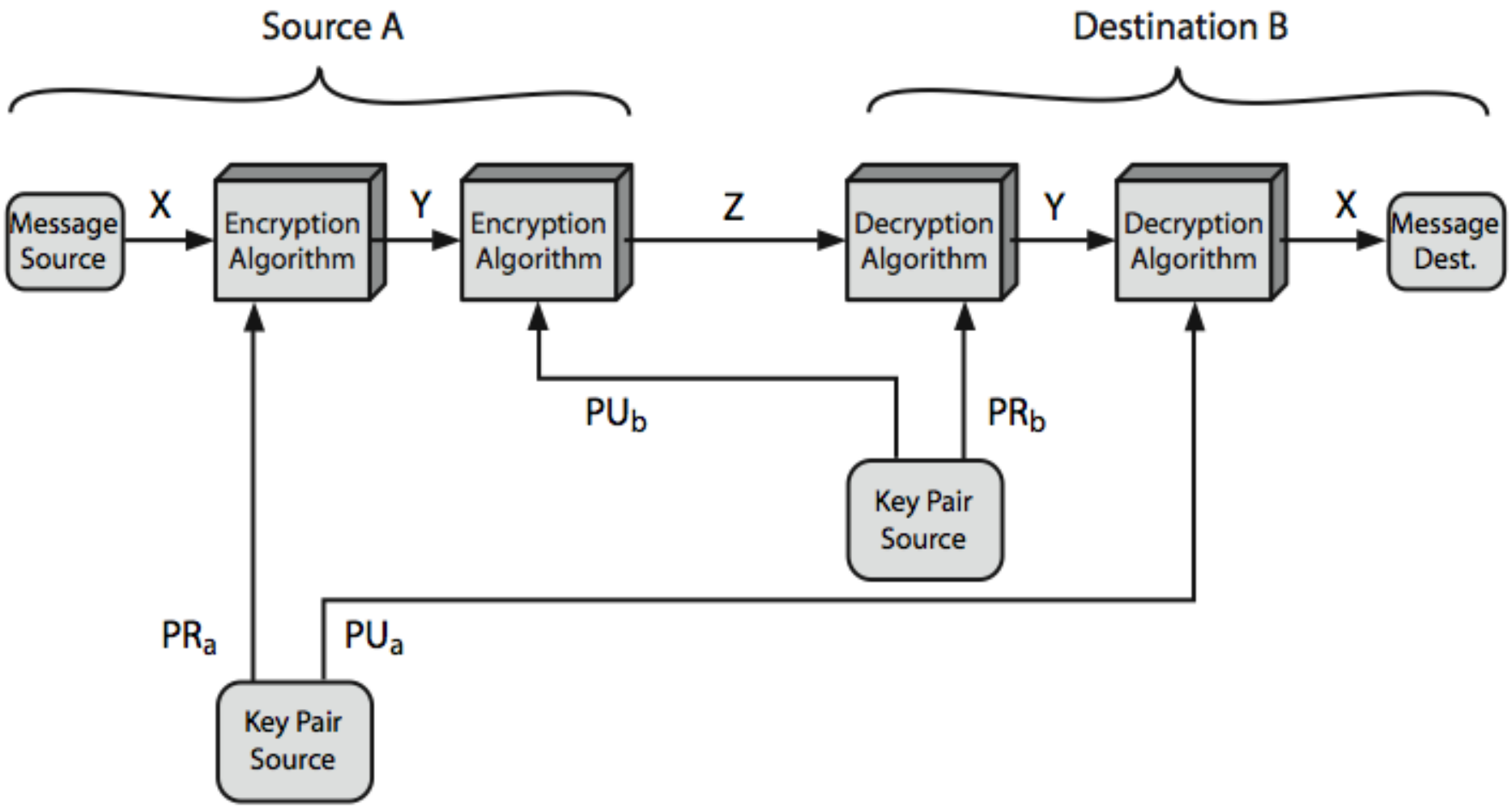
Asymmetric Encryption



PKC Authentication



PKC - Encryption & Authentication



Public Key characteristics

- Public-Key algorithms rely on two keys where:
  - it is computationally infeasible to find decryption key knowing only algorithm & encryption key
  - it is computationally easy to en/decrypt messages when the relevant (en/decrypt) key is known

PKC Applications

- can classify uses into 3 categories
  - encryption/decryption (provide secrecy)
    - the sender encrypts a message with the recipient's public key.
  - digital signature (provide authentication)
    - the sender "signs" a message with its private key.
  - key exchange (of session keys)
    - two sides cooperate to exhange a session key.
- some algorithms are suitable for all uses, others are specific to one

Public Key requirements

- Formally a PKC is a pair of families $\{E_k\}_{k \in \{K\}}$ and $\{D_k\}_{k \in \{K\}}$ of algorithms
  - representing invertible transformations on a finite message space such that,
    - for every $K \in \{K\}$, E_k is the inverse of D_k
    - it is computationally easy for a party B to generate a pair (public key $K_{PUb} \in \{K\}$, private key $K_{PRb} \in \{K\}$)

Public Key requirements (contd)

- for every **K** and **M**, the algorithms $\mathbf{E_k}$ and $\mathbf{D_k}$ are easy to compute such that
  - it is easy for sender a to generate $C = E_{PUb}(M)$
  - it is easy for the receiver b to compute $M = D_{PRb}(C) = D_{PRb}[E_{PUb}(M)]$